

AQUA SENTINEL — Drone Water Quality Survey

Location: Dal Lake, Srinagar, Jammu & Kashmir, India

Coordinates (centroid): 34.0837°N, 74.8600°E

Altitude: ~1,583 m ASL

Survey Date: 2026-05-05

Drone ID: AQ-SENTINEL-07

Sortie Window: 06:30 – 07:42 IST

Samples Collected: 25

Sensor Probe: HydroSense Multi-Parameter v3.1

1. Location & Environmental Context

Dal Lake is an urban freshwater lake situated in the heart of Srinagar city at an elevation of approximately 1,583 m above sea level. It spans roughly 18 km² and serves as a critical ecological, economic, and cultural resource for the Kashmir Valley. The lake receives inflow from multiple channels including Dachigam Nallah and Telbal Nallah, and has historically suffered from eutrophication driven by agricultural runoff, houseboat sewage, and encroachment. Seasonal algal blooms, elevated nutrient loading, and declining water transparency have been documented in numerous CSIR-IITR and LAWDA studies. The current survey was commissioned to provide a high-resolution snapshot of water quality parameters across the lake surface.

2. Sensor Specifications

Parameter	Sensor	Range	Accuracy	Unit
pH	ISE glass electrode	0 – 14	± 0.05	—
Turbidity	Nephelometric (IR 860 nm)	0 – 1000	± 2%	NTU
Dissolved Oxygen	Optical luminescence	0 – 20	± 0.1	mg/L
Temperature	Pt1000 RTD	-5 – 50	± 0.1	°C
Conductivity	4-electrode cell	0 – 2000	± 1%	µS/cm
Ammonia (NH3-N)	Ion-selective electrode	0 – 10	± 5%	mg/L
ORP	Pt/Ag-AgCl redox	-999 – +999	± 1	mV
GPS	u-blox NEO-M9N	—	± 1.5 m CEP	WGS84

3. Raw Sensor Readings

Each row represents a single dip-sample taken by the drone probe at the GPS coordinates shown. Readings are ordered chronologically.

ID	Timestamp	Lat	Lon	pH	Turb (NTU)	DO (mg/L)	Temp (°C)	Cond (µS/cm)	NH3 (mg/L)	ORP (mV)
DL-001	2026-05-05 06:30:00	34.0834	74.8597	7.76	22.21	6.35	15.25	334.95	0.41	203.5
DL-002	2026-05-05 06:33:00	34.0839	74.8605	8.21	21.94	6.63	16.39	274.34	0.487	249.3
DL-003	2026-05-05 06:36:00	34.0838	74.8598	7.99	9.28	6.13	18.24	359.3	0.414	221.3
DL-004	2026-05-05 06:39:00	34.0842	74.8616	7.41	21.41	4.68	13.57	292.69	0.313	236.3
DL-005	2026-05-05 06:42:00	34.0850	74.8576	8.1	11.99	6.4	17.06	325.15	0.573	229.2

ID	Timestamp	Lat	Lon	pH	Turb (NTU)	DO (mg/L)	Temp (°C)	Cond (µS/cm)	NH3 (mg/L)	ORP (mV)
DL-006	2026-05-05 06:45:00	34.0844	74.8613	7.97	14.24	5.64	16.8	342.47	0.412	280.1
DL-007	2026-05-05 06:48:00	34.0821	74.8578	8.07	26.53	7.11	18.75	384.19	0.436	167.3
DL-008	2026-05-05 06:51:00	34.0826	74.8619	7.29	18.2	6.8	17.03	352.56	0.537	279.6
DL-009	2026-05-05 06:54:00	34.0849	74.8588	7.6	13.01	7.64	16.65	316.84	0.562	188.3
DL-010	2026-05-05 06:57:00	34.0831	74.8563	7.42	14.59	7.0	19.29	319.17	0.489	215.0
DL-011	2026-05-05 07:00:00	34.0859	74.8618	7.9	11.93	7.58	18.07	375.21	0.446	268.6
DL-012	2026-05-05 07:03:00	34.0830	74.8632	7.84	14.9	5.15	17.27	384.05	0.572	230.7
DL-013	2026-05-05 07:06:00	34.0789	74.8614	7.99	14.7	5.75	17.5	397.62	0.292	197.2
DL-014	2026-05-05 07:09:00	34.0864	74.8591	7.67	18.59	5.01	17.83	265.57	0.583	210.1
DL-015	2026-05-05 07:12:00	34.0883	74.8606	8.28	10.18	6.35	17.98	398.56	0.198	239.7
DL-016	2026-05-05 07:15:00	34.0849	74.8631	8.05	18.31	5.87	15.63	328.79	0.421	270.6
DL-017	2026-05-05 07:18:00	34.0825	74.8606	7.25	15.63	6.81	18.74	385.16	0.443	176.5
DL-018	2026-05-05 07:21:00	34.0846	74.8610	7.97	13.8	7.86	17.63	351.49	0.641	228.3
DL-019	2026-05-05 07:24:00	34.0843	74.8643	7.89	16.22	6.63	19.72	325.34	0.528	245.9
DL-020	2026-05-05 07:27:00	34.0827	74.8565	7.9	19.38	5.77	18.81	347.34	0.301	225.5
DL-021	2026-05-05 07:30:00	34.0833	74.8570	7.96	17.73	5.61	18.28	341.96	0.363	222.0
DL-022	2026-05-05 07:33:00	34.0857	74.8586	7.93	18.02	9.22	14.71	351.29	0.404	207.0
DL-023	2026-05-05 07:36:00	34.0875	74.8600	8.58	15.35	6.92	16.76	288.92	0.459	201.5
DL-024	2026-05-05 07:39:00	34.0841	74.8558	8.5	17.0	8.59	15.99	333.16	0.925	183.9
DL-025	2026-05-05 07:42:00	34.0807	74.8589	7.97	22.16	8.05	17.11	247.9	0.385	247.2

4. Summary Statistics

Parameter	Min	Max	Mean	Std Dev
pH	7.250	8.580	7.900	0.324
Turbidity (NTU)	9.280	26.530	16.692	4.052
DO (mg/L)	4.680	9.220	6.622	1.090
Temperature (°C)	13.570	19.720	17.242	1.426
Conductivity (µS/cm)	247.900	398.560	336.961	39.663
Ammonia (mg/L)	0.198	0.925	0.464	0.139
ORP (mV)	167.300	280.100	224.984	30.395

5. Applicable Regulatory Benchmarks

The following standards are referenced for comparison: CPCB (Central Pollution Control Board) primary water quality criteria, BIS IS 10500:2012 (drinking water — for reference only, Dal Lake is not a potable source), and WHO Guidelines for Recreational Water Quality.

Parameter	CPCB Class B (Bathing)	BIS 10500 (Drinking)	WHO (Recreation)
pH	6.5 – 8.5	6.5 – 8.5	6.5 – 8.5
Turbidity (NTU)	—	≤ 5	—

Parameter	CPCB Class B (Bathing)	BIS 10500 (Drinking)	WHO (Recreation)
DO (mg/L)	≥ 3	—	—
Temperature (°C)	—	—	—
Conductivity (μS/cm)	—	—	—
Ammonia (mg/L)	—	≤ 0.5	—
ORP (mV)	—	—	≥ 200 (ideal)

6. Field Notes

- Weather: Clear skies, wind < 5 km/h from NW. Air temperature 14 °C at sortie start.
- Visible algal mats observed near southern shore (Nehru Park side).
- Houseboat cluster in NE quadrant — potential point-source contamination.
- Probe was calibrated on-site using pH 4.01 / 7.00 / 10.01 buffers and zero-DO solution.
- Data transmitted in real-time via 4G LTE; no telemetry gaps recorded.

Report generated automatically by Aqua Sentinel data pipeline. For queries contact: aqua-sentinel@droneops.in